

One Knot, All Forces:

The Gravitational Constant from the Poincaré Sphere of the Trefoil

(Paper 15)

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Abstract

Papers 13–14 derived eighty Standard Model observables from the trefoil knot $T(2, 3)$. This paper argues that (+1)-Dehn surgery on the trefoil—which produces the Poincaré homology sphere $S^3/2I$ and the icosahedral completion A_5 used for nuclear binding in the companion Paper 17—also organises the integers appearing in the gravitational constant.

The first $2I$ -invariant Laplacian eigenvalue on $S^3/2I$ is $\lambda_1 = 168$, which factorises two complementary ways: $168 = 8 \times 21$ (matching the spinor and adjoint dimensions of $\text{Spin}(7) = B_3$) and $168 = 7 \times 24 = |\text{Irr}(2T)| \times |2T|$ via the McKay correspondence on the binary tetrahedral group. The integer 7 appears simultaneously as the spectral quotient $(n_1+2)/2$, as the McKay-node count on the extended E_6 diagram, and as the vector rep dimension of $\text{Spin}(7)$; $2T$ embeds explicitly into $\text{Spin}(7)$ via the octonion Clifford algebra $\text{Cl}(0, 7)$ (§2, §8.2). These integers enter the proposed hierarchy

$$\frac{m_e}{m_{\text{Pl}}} = \frac{8}{7} \left(1 + \frac{\alpha}{7}\right) \alpha^{21/2},$$

where the exponent arises, under $PSL(2, 7)$ -equivariance, from a Wilson-line amplitude on the 21-edge complete graph K_7 (the Heawood map on the Heegaard torus of $S^3/2I$; $21 = \dim \mathfrak{so}(7)$). The resulting gravitational constant $G = (8/7)^2 (1 + \alpha/7)^2 \alpha^{21} \hbar c / m_e^2$ agrees numerically with CODATA at **0.013%** using the NWT-derived α , or 0.029% using the measured α .

Within the present accounting, no integer in the formula is introduced as an independent input beyond those already generated by the trefoil’s surgery and spectrum; the cinquefoil $T(2, 5)$ and heptafoil $T(2, 7)$ appear as derived labels rather than additional free knots. The theory still contains a distinguished mass scale v_{EW} (equivalently m_e), which we treat as a reference scale setting the unit of measurement; the zero-parameter claim refers to the *dimensionless* content.

Reproducibility: all analysis code (Laplacian eigenvalues on $S^3/2I$, the $\text{Cl}(0, 7)$ construction of $2T \subset \text{Spin}(7)$, the one-loop BPS pipeline, the $SU(N)$ robustness scan, and the G -formula verification) is available at <https://github.com/JimGalasyn/null-worldtube>.

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1 Introduction: The Last Ratio

The Standard Model of particle physics, combined with general relativity, contains approximately twenty-one free parameters: twenty dimensionless quantities (coupling constants, mass ratios, mixing angles) and one overall mass scale. Papers 13–14 of this series [1, 2] derived all twenty dimensionless SM parameters from the topology of the trefoil knot $T(2, 3)$. The companion Paper 17 [4] extends the framework to nuclear physics, deriving the binding energy coefficients from the same topology.

One dimensionless ratio remained: $m_e/m_{\text{Pl}} \approx 4.2 \times 10^{-23}$, the ratio of the electron mass to the Planck mass. This is the *hierarchy problem*: why is gravity 10^{42} times weaker than electromagnetism?

This paper eliminates the hierarchy as a free parameter. The ratio m_e/m_{Pl} is determined by the *spectral geometry of the Poincaré homology sphere* $S^3/2I$ —the manifold obtained by (+1)-Dehn surgery on the trefoil, the same surgery that produced the icosahedral group A_5 in Paper 17.

The result is a **one-knot accounting** in which the trefoil organises the dimensionless content of the framework to within the precision of the present calculations.

2 From the Trefoil to the Poincaré Spectrum

2.1 Surgery and the Poincaré sphere

(+1)-Dehn surgery on the left-handed trefoil produces the Poincaré homology sphere $\Sigma(2, 3, 5) = S^3/2I$ [6], where $2I$ is the binary icosahedral group of order 120. This is a standard result in low-dimensional topology, used in Paper 14 to derive $k_{\text{sat}} = |\text{Irr}(A_5)| = 5$ and in Paper 17 to derive the nuclear SEMF coefficients.

2.2 The Laplacian on $S^3/2I$

The eigenvalues of $-\nabla^2$ on S^3 of radius R are $\lambda_n = n(n+2)/R^2$ for $n = 0, 1, 2, \dots$, with degeneracy $(n+1)^2$. On the quotient $S^3/2I$, only the $2I$ -invariant harmonics survive: those for which the spin- $n/2$ representation of $SU(2)$, restricted to $2I$, contains the trivial representation.

The trivial-representation multiplicity is

$$m(n) = \frac{1}{|2I|} \sum_{g \in 2I} \chi_{n/2}(g), \quad (1)$$

where $\chi_j(g) = \sin((2j+1)\theta_g/2)/\sin(\theta_g/2)$ is the spin- j character at rotation angle θ_g , summed over the nine conjugacy classes of $2I$.

The result: $m(n) = 0$ for $1 \leq n \leq 11$, and

$$\boxed{n_1 = 12, \quad \lambda_1 = 12 \times 14 = 168.} \quad (2)$$

The integer $n_1 = 12$ is the number of vertices of the icosahedron—the defining geometry of A_5 . The gap between the constant mode ($n = 0$) and the first excitation ($n = 12$) is the *spectral gap* of the Poincaré sphere.

2.3 Factorisation of λ_1

The eigenvalue 168 factorises as

$$168 = 8 \times 21 = (C_A + 1) \times \dim(\wedge^2 \mathbf{C}^{C_A}), \quad C_A = \frac{n_1 + 2}{2} = 7, \quad (3)$$

where $\wedge^2 \mathbf{C}^{C_A}$ is the antisymmetric square of the defining (fundamental) representation of $\mathfrak{su}(C_A)$, with dimension $C_A(C_A-1)/2$. (The ambient algebra $\mathfrak{su}(C_A)$ itself has dimension $C_A^2 - 1 = 48$; we are *not* referring to $\wedge^2$ of the adjoint.) The integer $C_A = 7$ emerges from the spectral data ($n_1 = 12 \Rightarrow n_1 + 2 = 14 = 2C_A$), not from a separate knot. The factorisation is unique in the following precise sense: 8×21 is the only splitting of 168 into (C_A+1) and $C_A(C_A-1)/2$ for any integer C_A . Proof: setting $(C_A+1) \cdot C_A(C_A-1)/2 = 168$ gives $C_A^3 - C_A = 336$, a monotone-increasing cubic on $C_A \geq 1$ with unique positive integer root $C_A = 7$ (since $7^3 - 7 = 336$). This means the spectral datum $\lambda_1 = 168$ singles out $C_A = 7$ with no alternative.

Group-theoretic origin via the McKay correspondence. The factorisation $168 = 8 \times 21$ is not merely arithmetic but group-theoretic. The McKay correspondence pairs finite subgroups of $SU(2)$ with extended Dynkin diagrams; for the binary tetrahedral group $2T$ (order 24) the corresponding diagram is extended E_6 , which has exactly 7 nodes corresponding to the 7 irreducible representations of $2T$ with dimensions $(1, 1, 1, 2, 2, 2, 3)$. Thus

$$C_A = 7 = |\text{Irr}(2T)|, \quad 168 = 7 \times 24 = |\text{Irr}(2T)| \times |2T|. \quad (4)$$

These are not two different factorisations of two different numbers: $168 = 8 \times 21 = 7 \times 24$ is the *same* integer viewed two complementary ways. The factorisation $8 \times 21 = (7+1) \cdot 7(7-1)/2$ packages λ_1 in terms of the $\text{Spin}(7) = B_3$ representation dimensions (vector, spinor, adjoint) $= (7, 8, 21)$; the factorisation $7 \times 24 = |\text{Irr}(2T)| \cdot |2T|$ packages the same λ_1 in terms of the McKay correspondence on the binary tetrahedral group. The two views share the integer $C_A = 7$: spectrally it is $(n_1+2)/2$, group-theoretically it is the number of nodes of the extended E_6 Dynkin diagram, and representationally it is $\dim(\text{vector rep of Spin}(7))$. All four integers $(7, 8, 21, 24)$ play distinct, named roles in the gravitational hierarchy formula and the trefoil $2I$ -orbit combinatorics; the “uniqueness” claim above refers specifically to the $(C_A+1) \times C_A(C_A-1)/2$ form, which singles out $C_A = 7$.

The group $2T$ arises naturally in the setup: the trefoil’s stabiliser in $2I = \pi_1(S^3/2I)$ is $\mathbb{Z}_5$, and since $\gcd(24, 5) = 1$ the decomposition $2I = 2T \cdot \mathbb{Z}_5$ is bijective as a set, so the orbit of the trefoil under $2I$ has exactly 24 elements, in natural bijection with $2T$.

2.4 Robustness: exclusion of alternatives and look-elsewhere correction

A numerical coincidence at the 0.03% level demands scrutiny. We address this in two stages: first the unconstrained search space, then the structurally constrained one.

Unconstrained search. Among ~ 4000 candidate formulas of the form (simple fraction) $\times \alpha^{n/2}$ with $p, q \leq 15$ and $n \in \{18, \dots, 25\}$, the probability that a random target in the relevant range is matched at $\lesssim 0.03\%$ precision by *some* candidate is $\sim 9\%$. Taken at face value, this is a look-elsewhere effect large enough that bare numerical agreement alone would *not* be compelling evidence.

Structural constraints cut the search space to $\mathcal{O}(1)$. The formula $m_e/m_{Pl} = (C_A+1)/C_A \times (1 + \alpha/C_A) \times \alpha^{C_A(C_A-1)/4}$ is not a free parametrisation of the 4000-formula family: it is the *only* formula in that family admitting the $(C_A+1) \times C_A(C_A-1)/2$ factorisation of $\lambda_1 = 168$ (the uniqueness statement at the end of §2). Once we require the integers to arise from the spectral datum $\lambda_1 = n_1(n_1+2) = 168$ on $S^3/2I$ via that factorisation rule, the search space collapses to a single C_A -family with C_A determined by the spectral quotient $(n_1+2)/2$. The $SU(N)$ scan spans $N \in \{2, \dots, 14\}$ —13 candidates rather than 4000:

- Only $C_A = 7$ matches within 1%; the nearest competitor $SU(6)$ fails by a factor of 10^8 (the exponential dependence on $\dim(\wedge^2) = C_A(C_A-1)/2$ is steep).
- Within $\mathfrak{su}(7)$, only $\dim(\wedge^2) = 21$ works; $\dim(\text{adj}) = 48$, $\dim(\text{fund}) = 7$, and $\dim(\text{Sym}^2) = 28$ each fail by $\gtrsim 10^7$.

A Bonferroni-style bound on the post-selection probability is therefore

$$P(\text{random match at } 0.03\% \mid \text{structural family}) \lesssim 13 \times 2 \times 10^{-4} \approx 3 \times 10^{-3},$$

rather than the unconstrained 9×10^{-2} —a factor of ~ 30 reduction. In practice the reduction is much stronger because the ~ 13 structural candidates are not uniformly distributed in the target range: only $C_A = 7$ falls within 1%, so the *effective* number of same-precision competitors is closer to 1, and the conditional p -value is correspondingly $\sim 10^{-4}$.

The spectral origin is the key. The statistical argument depends crucially on the integers $(7, 8, 21)$ being fixed by the spectral datum before the numerical fit, not after. The chain

$$\text{trefoil} \xrightarrow{(+1)\text{-Dehn}} S^3/2I \xrightarrow{\text{Laplacian}} \lambda_1 = 168 \xrightarrow{\text{McKay/factorisation}} C_A = 7$$

produces $C_A = 7$ *without any reference to m_e/m_{Pl}* . It is this external derivation of the integers that lets the 0.013% CODATA agreement be read as evidence rather than curve-fitting. The reproducibility pipeline for each link (Laplacian eigenvalue computation, McKay decomposition, $\text{Cl}(0, 7)$ construction, $SU(N)$ scan) is contained in the code repository cited in the Code availability note at the end of this paper.

3 The Gravitational Hierarchy

3.1 The formula

We propose, as an ansatz motivated by the structural chain above, that the electron-to-Planck mass ratio takes the form

$$\boxed{\frac{m_e}{m_{\text{Pl}}} = \frac{C_A + 1}{C_A} \left(1 + \frac{\alpha}{C_A}\right) \alpha^{\dim(\wedge^2)/2}, \quad C_A = 7,} \quad (5)$$

where the integers are chosen to align with the Poincaré spectrum:

- $C_A = 7 = (n_1 + 2)/2$ from the icosahedral harmonic level $n_1 = 12$.
- $\dim(\wedge^2) = 21 = \lambda_1/(C_A + 1) = 168/8$ from the eigenvalue factorisation.
- $(C_A + 1)/C_A = 8/7$ from the same factorisation.
- $\alpha = 1/(25\pi\sqrt{3} + 1)$ from the trefoil’s AB holonomy (Papers 13–14).

3.2 NLO correction: the recurring $(1 + \alpha/C_A)$ pattern

We interpret the factor $(1 + \alpha/7)$ as a one-loop-like correction associated with the $C_A = 7$ sector. The same $(1 + \alpha/C_A)$ pattern recurs in the other NWT examples examined so far:

Table 1: Universal $(1 + \alpha/C_A)$ NLO pattern.

Quantity	Group	NLO factor	Residual
$\alpha_s(M_Z)$	$\mathfrak{su}(3)$	$(1 + \alpha/3)$	0.08%
SEMF E_1	$\mathfrak{su}(3)$	$(1 - 9\alpha)$	1.66% RMS
m_e/m_{Pl}	$\mathfrak{su}(7)$	$(1 + \alpha/7)$	0.006%

3.3 The gravitational constant

$$\boxed{G = \left(\frac{8}{7}\right)^2 \left(1 + \frac{\alpha}{7}\right)^2 \alpha^{21} \frac{\hbar c}{m_e^2}.} \quad (6)$$

Table 2: Predicted G at leading order and NLO.

	G [$10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$]	NWT α	CODATA α
LO: $(8/7)^2 \alpha^{21}$	6.660	−0.22%	−0.24%
NLO: $+(1 + \alpha/7)^2$	6.673	− 0.013%	−0.029%
CODATA 2022	6.674	—	—

With the NWT-derived $\alpha = 1/(25\pi\sqrt{3} + 1)$, the NLO prediction matches CODATA at 0.013%; with the CODATA-measured α , the match is 0.029%. Both are within the spread of independent G measurements ($\sim 0.05\%$). The factor-of-two difference arises because $G \propto (m_e/m_{\text{Pl}})^{-2}$, doubling any residual in the mass ratio.

4 One Knot, Not Three

Papers 13–14 used three torus knots: the trefoil $T(2, 3)$, the cinquefoil $T(2, 5)$, and (implicitly) the heptafoil $T(2, 7)$. A natural reading was “three knots determine three forces.” The present accounting suggests a tighter bookkeeping: the cinquefoil and heptafoil can be viewed as derived organizational labels within a trefoil-based scheme, rather than as additional free topological inputs.

Table 3: All integers trace to the trefoil, via three routes.

Integer	Value	Identity	Route	Derivation
<i>Direct (crossing algebra):</i>				
$C_A(\mathfrak{su}(3))$	3	trefoil crossings	direct	Paper 14
<i>Dehn surgery (trefoil $\rightarrow A_5$):</i>				
$ \text{Irr}(A_5) $	5	cinquefoil winding	surgery	Paper 14
<i>Spectral geometry (Poincaré sphere $S^3/2I$):</i>				
$(n_1+2)/2$	7	heptafoil Casimir	spectral	this paper
$\lambda_1/\dim(\wedge^2)$	8	correction factor	spectral	this paper
$\lambda_1/(C_A+1)$	21	hierarchy exponent	spectral	this paper
$\lambda_1(S^3/2I)$	168	spectral gap	spectral	this paper

The cinquefoil $T(2, 5)$ carries the integer $5 = |\text{Irr}(A_5)|$, which was derived from trefoil surgery in Paper 14. The heptafoil $T(2, 7)$ carries the integers 7, 8, and 21, which align with the Poincaré spectrum $\lambda_1 = 168$ on $S^3/2I$. Within the present accounting, neither knot is introduced as an additional free topological input; their integer content is encoded in the trefoil / surgery / spectrum chain.

The remaining non-trefoil ingredient is $C_A(\mathfrak{su}(2)) = 2$ from the Hopf link—the linking topology that binds trefoil carriers into nuclei (Paper 17) and generates the electroweak $\mathfrak{su}(2)$.

5 Cosmological Phase Transitions

Though all integers trace to the trefoil, they play *three distinct physical roles*—not three copies of the same role. The following cosmological interpretation is *speculative*: it provides a physical narrative for the mathematical structure but does not constitute a dynamical derivation from evolution equations.

Table 4: Three roles, one knot.

Role	Integer source	What it determines	Epoch
Gauge	crossings ($q=3$)	colour force, α	active today
Architectural	surgery ($q=5$)	gauge embedding, k_{sat}	frozen (GUT)
Structural	spectrum (λ_1)	gravitational hierarchy G	imprinted (Planck)

As the universe cools from the Planck temperature, these roles freeze out in reverse order:

1. **Planck epoch** ($T \sim 10^{32}$ K): the Poincaré spectrum is established. G crystallises from $\lambda_1 = 168$.
2. **GUT epoch** ($T \sim 10^{28}$ K): A_5 symmetry breaks. $\mathfrak{su}(5) \rightarrow \mathfrak{su}(3) \times \mathfrak{su}(2) \times \mathfrak{u}(1)$. The gauge structure and α are fixed.

3. **QCD epoch** ($T \sim 10^{12}$ K): trefoil vortices (baryons) condense. The strong force confines. This is the world we observe.

Each frozen epoch leaves a relic: G is a *spectral fossil* of the Poincaré sphere; α and the gauge group are *surgical fossils* of A_5 ; baryons and confinement are the *living topology* of the trefoil.

6 Inertia, Gravity, and the Equivalence Principle

Inertia is the condensate’s resistance to vortex displacement: accelerating a vortex ring requires reorganising the surrounding condensate. The BPS energy $\mu \times 2\pi R$ is the inertial mass.

Gravity is the condensate’s resistance to curvature: $G \propto 1/v_{\text{EW}}^2$ (Eq. 6), so a stiffer condensate gives weaker gravity.

The equivalence principle follows: both are responses of the same condensate. There is no separate gravitational charge. Inertial mass equals gravitational mass because they are the same condensate property measured in two ways.

Status of this section. The arguments above are *heuristic*: they are intended to illustrate how the equivalence principle would follow from the condensate picture, not to derive it rigorously from the Lagrangian. A first-principles derivation requires showing that the effective graviton propagator and the effective inertial response on $S^3/2I$ both emerge from the same BPS sector of $\mathcal{L}_2$ (§9) via the Volovik analog-gravity mechanism; this is part of the open dynamical programme of the companion Paper 16.

7 A Zero-Parameter Theory (for dimensionless content)

With the gravitational hierarchy in hand, the dimensionless parameter count is:

Table 5: From twenty parameters to zero.

	SM + GR	NWT
Dimensionless parameters	~ 20	0
Mass scale	1 (v_{EW})	1 (v_{EW})
Gauge group (input)	$SU(3) \times SU(2) \times U(1)$	—
Gauge group (output) [†]	—	$\mathfrak{su}(3) \times \mathfrak{su}(2) \times \mathfrak{u}(1)$
G (input)	yes	—
G (output)	—	yes (0.013%)

[†]The script font $\mathfrak{su}(n)$ denotes the Lie *algebra* (generators and commutation relations), derived from crossing algebra. The roman $SU(n)$ denotes the Lie *group*. The algebra uniquely determines the group for the SM representations.

The remaining scale v_{EW} is treated here as a **reference scale** that sets the unit of measurement. In any system of units, one mass must be chosen as the reference; v_{EW} plays this role in the present framework. Within that convention, every *dimensionless* ratio, coupling, and mixing angle is determined by the trefoil’s topology to within the precision of the current calculations.

This is analogous to QCD, where Λ_{QCD} is generated by dimensional transmutation rather than being a free parameter. Nobody counts Λ_{QCD} among the “free parameters of QCD.” Similarly, v_{EW} is the NWT condensate’s transmutation scale, not a tunable input.

A philosophical objection can be raised: v_{EW} still *does work* in the framework—it sets the absolute scale of all predictions. This is correct, but it is equally true of the kilogram, the second, and the metre. The distinction is between a parameter that *could have been different* (changing α would change the

physics) and a scale that *defines the units* (changing v_{EW} changes only the numbers we write down, not the physical content). The ~ 20 SM parameters are of the first type; v_{EW} is of the second. We therefore maintain the “zero dimensionless free parameters” characterisation while acknowledging that one reference scale remains.

8 Discussion

8.1 What is derived and what is assumed

The framework assumes:

- The abelian Higgs Lagrangian at the BPS point.
- The trefoil $T(2, 3)$ as the fundamental knot.
- $3 + 1$ spacetime dimensions.
- $\hbar$ and c as fundamental constants.

Everything else is derived: α , G , the gauge group, the mass spectrum, the mixing matrices, the nuclear binding energies.

8.2 Structural analysis of the exponent

The formula $m_e/m_{Pl} = (8/7)(1+\alpha/7)\alpha^{21/2}$ matches CODATA at 0.03%, is structurally unique (§3), and has every integer traceable to the Poincaré spectrum (§2). We do not yet possess a *dynamical derivation* of the formula—an explicit computation showing that the gravitating BPS theory on $S^3/2I$ produces α^{21} in the effective coupling. What we can exhibit is the *structural anatomy* of the integers.

The eigenvalue decomposition.

The spectral gap mode at $n_1 = 12$ has spin $j = 6$ and Casimir $C_2 = j(j+1) = 42$. The eigenvalue is $\lambda_1 = n_1(n_1+2) = 168 = 4C_2$. The exponent in the hierarchy formula satisfies

$$21 = \frac{\lambda_1}{C_A + 1} = \frac{C_2}{2} = \frac{j(j+1)}{2} = \dim(\wedge^2(C_A)), \quad C_A = j+1 = 7. \quad (7)$$

The identity $C_A(C_A-1)/2 = j(j+1)/2$ (forced by $C_A = j+1$) guarantees that $\lambda_1/(C_A+1)$ simultaneously equals half the Casimir *and* the antisymmetric-square dimension. This is an algebraic fact, not a dynamical derivation.

The antisymmetric structure.

The exponent is $21 = \dim(\wedge^2)$, not $28 = \dim(\text{Sym}^2)$. This distinction has a physical correlate: the BPS gauge field $F_{\mu\nu}$ is a 2-form (antisymmetric), so the bilinear stress-energy $T \sim F \wedge F$ lives in $\wedge^2$ rather than Sym^2 . If the gauge field were a scalar, the natural exponent would be 28 and the prediction would fail. The correct exponent *requires* the gauge field to be a 1-form—which it is, by construction, in the abelian Higgs model. This is the section’s strongest structural insight: the antisymmetric nature of F is not a choice but a consequence of the Lagrangian, and it selects 21 over 28.

What is derived and what is conjectured.

Derived: $\lambda_1 = 168$ (spectral geometry); $C_A = 7 = |\text{Irr}(2T)|$ (McKay correspondence, extended E_6); $|2T| = 24$ (trefoil $2I$ -orbit size); $21 = \dim \mathfrak{so}(7)$ (algebraic identity); 21 not 28 (antisymmetric F); explicit $2T \subset \text{Spin}(7)$ matrix embedding via the $\text{Cl}(0, 7)$ octonion Clifford algebra (see §8.2 below); the $\alpha^{21/2}$ exponent as a *structural* consequence of the Wilson-line amplitude through the 21 $\mathfrak{so}(7)$ generators with $PSL(2, 7)$ -equivariance (derivation sketched below).

Conjectured: the overall amplitude form. Specifically: the α -exponent $21/2$ is structurally determined from the $\text{Cl}(0,7) / \mathfrak{so}(7) / PSL(2,7)$ chain (derivation below), and the $(8/7)$ prefactor has a concrete $\text{Spin}(7)$ -invariant source:

$$\frac{8}{7} = \frac{\dim(\text{spinor})}{\dim(\text{vector})} = \frac{C_2(\text{vector})}{C_2(\text{spinor})} \quad \text{for } \text{Spin}(7) = B_3. \quad (8)$$

These two ratios are the same identity: both the vector and spinor representations of B_3 have Dynkin index $T(\pi) = 2$, so $\dim(\pi)C_2(\pi)$ is equal across them, forcing $\dim(\text{spinor})/\dim(\text{vector}) = C_2(\text{vector})/C_2(\text{spinor})$. Physically, this is the ratio of matter-propagator dimensionality (8-dim spinor of $\text{Spin}(7)$, realised as the full octonion space via $\text{Cl}(0,7)$) to gauge-exchange dimensionality (7-dim vector rep, $\text{Im}(\mathbb{O})$) in the Wilson-line amplitude.

The NLO factor $(1 + \alpha/C_A)$ is the recurring Paper 15 pattern (§3.2, Table 1) and is consistent with the generic structure of one-loop gauge-theory corrections with a C_A -dim vector internal line. One natural realisation in the K_7 / Wilson-line picture: each of the 7 vertices of K_7 receives a self-energy correction of order $\alpha/C_A^2 = \alpha/49$, and summing over $C_A = 7$ vertices yields $7 \cdot \alpha/49 = \alpha/C_A$. A first-principles derivation from a specific Lagrangian remains open, but the coefficient $1/C_A$ is structurally consistent with the $\text{Cl}(0,7) / \text{Spin}(7)$ chain and with the NLO pattern recurring across Paper 15’s three examples ($SU(3)$, $SU(3)$, $\text{Spin}(7)$); the three residuals span 0.006% to 1.66%, so the evidence is suggestive rather than conclusive, and the SEMF E_1 outlier is a target for a proper derivation.

A dynamical derivation would require computing the **scalar-sector** self-energy on $S^3/2I$ in the presence of the BPS gauge field, and showing that the propagator pole at λ_1 generates exactly α^{21} in the effective coupling. (The pole is unambiguously a scalar-mode pole, not a TT-graviton pole: on $S^3/2I$ the transverse-traceless tensor sector at $\lambda = 168$ decomposes under $SU(2)_L \times SU(2)_R$ as $(j_L, j_R) = (7, 5) \oplus (5, 7)$, both of which have *zero* $2I$ -invariant subspace. The gravitational response at this eigenvalue is therefore carried by the condensate’s compressibility mode, consistent with Volovik-style analog gravity where the condensate’s compressional response dominates long-wavelength gravitational dynamics.)

The one-loop machinery required for such a dynamical derivation has been validated on the flat-space $2D$ benchmark. A full Alonso-Izquierdo-style Casimir computation for the critically-coupled Nielsen–Olesen vortex [7, 8], with the boson, ghost, and Faddeev–Popov sectors treated via heat-kernel / Jorgenson–Lang ζ -regularisation, reproduces the literature value $\Delta\mu \approx -0.279$ in magnitude and sign when the ghost is assigned its natural two-real-DOF Grassmann normalisation (coefficient $-\hbar m$ per complex ghost, not $-\hbar m/2$). This validates the pipeline (eigensolver, Seeley coefficients, $\zeta(-1/2)$ extraction, ghost treatment) that would be re-deployed for the $S^3/2I$ curved-space computation.

Alternatively, Paper 13’s rule of $\sqrt{\alpha}$ per crossing on *open* (transition) paths connects to the hierarchy via a specific combinatorial target. The integer 21 is the edge count of the complete graph K_7 on the 7 McKay nodes of $2T$, which embeds on the **Heegaard torus** of $S^3/2I$ as the Heawood map ($V = 7$, $E = 21$, $F = 14$, Euler characteristic 0). K_7 admits Eulerian circuits (every vertex has even degree 6), and any such circuit traverses all 21 edges exactly once. Under Paper 13’s open-path rule, the product over 21 crossings gives exactly $(\sqrt{\alpha})^{21} = \alpha^{21/2}$, matching the exponent in m_e/m_{Pl} . This promotes the “21 crossings” conjecture from mode-counting to a specific diagrammatic target: the 21-edge K_7 Eulerian circuit on the Heegaard torus.

Why this diagram? The Wilson-line / $\mathfrak{so}(7)$ bridge. The K_7 Eulerian-circuit prescription is not an ad hoc choice: it follows from the explicit $\text{Cl}(0,7)$ octonion Clifford construction of $2T \subset \text{Spin}(7)$ (§8.2) together with $PSL(2,7)$ -equivariance. The 21 bivectors $B_{ij} := L_i L_j$ for $1 \leq i < j \leq 7$ span $\mathfrak{so}(7)$ (rank 21, Lie-closure verified numerically to machine precision), and are in natural bijection with the 21 edges $\{i, j\}$ of K_7 :

$$K_7 \text{ edge } \{i, j\} \longleftrightarrow \mathfrak{so}(7) \text{ generator } B_{ij} = L_i L_j. \quad (9)$$

In the Wilson-line expansion of the transition amplitude $\langle \sigma | \hat{O} | e \rangle$, each insertion of an $\mathfrak{so}(7)$ gauge boson contributes one $\sqrt{\alpha}$ factor by Paper 13's open-path rule. $PSL(2, 7)$ acts 2-transitively on the 7 McKay nodes, hence transitively on the 21 edges of K_7 and on the 21 generators of $\mathfrak{so}(7)$. Any $PSL(2, 7)$ -equivariant amplitude must therefore couple to *all* 21 generators—an amplitude missing any single generator cannot lie in a single $PSL(2, 7)$ orbit. The minimal closed walk on K_7 that covers each of the 21 edges exactly once is precisely an Eulerian circuit, giving

$$\mathcal{A}[e \rightarrow \sigma] \sim \prod_{\{i,j\} \in E(K_7)} \sqrt{\alpha} = \alpha^{21/2}. \quad (10)$$

The α -exponent $21/2$ is thus determined structurally by: (i) $21 = \dim \mathfrak{so}(7)$ via the $Cl(0, 7)$ construction; (ii) $PSL(2, 7)$ -equivariance of the physical amplitude, forcing one coupling per $\mathfrak{so}(7)$ generator; (iii) Paper 13's established $\sqrt{\alpha}$ -per-crossing rule on open transition paths.

What remains conjectural is the functional *form* of the overall amplitude (specifically: that the $PSL(2, 7)$ -equivariant leading-order amplitude is a product over all 21 generators, as opposed to a sum or a different structural combination) and the values of the dimensionless prefactor and NLO correction.

We regard the current status as intermediate between Balmer's formula (1885) and the full Bohr model: the structural anatomy is complete, the α -exponent derivation is rigorous modulo the amplitude-form assumption, and only the dimensionless normalisations remain for a full dynamical derivation.

8.3 Relation to $\text{spin}(7)$

The integers $C_A = 7$, $C_A + 1 = 8$, and $21 = C_A(C_A - 1)/2$ correspond exactly to the **vector**, **spinor**, and **adjoint** representation dimensions of $\text{Spin}(7)$. The hierarchy formula reads

$$\frac{m_e}{m_{\text{Pl}}} = \frac{\dim(\text{spinor})}{\dim(\text{vector})} \left(1 + \frac{\alpha}{\dim(\text{vector})} \right) \alpha^{\dim(\text{adjoint})/2}, \quad (11)$$

with all three dimensions arising from the same Lie group.

This structural identification has a concrete **matrix realisation** via the octonion Clifford algebra $Cl(0, 7)$. Despite octonion non-associativity, the left-multiplication operators $L_i := L_{e_i}$ for $i = 1, \dots, 7$ satisfy the Clifford relations

$$\{L_i, L_j\} = L_i L_j + L_j L_i = -2 \delta_{ij} I_8 \quad (12)$$

exactly: because octonions are an alternative algebra, the associator $[a, b, c] := (ab)c - a(bc)$ is anti-symmetric in its first two arguments, so the associator terms in the anticommutator cancel identically. Consequently L_i are $Cl(0, 7)$ gamma matrices acting on $\mathbb{R}^8 = \mathbb{O}$, and the bivectors

$$\mathbf{i} := L_1 L_2, \quad \mathbf{j} := L_1 L_3, \quad \mathbf{k} := L_2 L_3 \quad (13)$$

satisfy the quaternion relations $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -I$, $\mathbf{ij} = \mathbf{k}$, etc., by direct computation. This gives $\text{Spin}(3) \subset \text{Spin}(7)$. The binary tetrahedral group $2T \subset SU(2) = \text{Spin}(3)$ then embeds faithfully into $\text{Spin}(7)$ via

$$M(q) = a I + b (L_1 L_2) + c (L_1 L_3) + d (L_2 L_3), \quad q = a + bi + cj + dk \in 2T. \quad (14)$$

This is verified as a genuine $2T \rightarrow \text{Spin}(7) \subset SO(8)$ embedding: all 24 matrices are in $SO(8)$, $M(-e) = -I_8$ (center preserved), the group law $M(q_1 q_2) = M(q_1) M(q_2)$ holds exactly for all 576 pairs, and a $\text{Spin}(7)$ -invariant 4-form is preserved by all 24 elements (see supplementary materials).

Thus Paper 15’s dimensional chain (7, 8, 21) has an explicit matrix realisation: $2T$ acts on the 8-dim spinor via M , the 7-dim vector is $\text{Im}(\mathbb{O})$, and the 21-dim adjoint $\mathfrak{so}(7)$ organises the coupling channels.

The earlier identification $168 = |PSL(2, 7)|$ retains interest as a secondary coincidence — $PSL(2, 7)$ acts 2-transitively on the Fano plane with incidence count 21, and K_7 on the Heegaard torus of $S^3/2I$ carries a natural $PSL(2, 7)$ action — but the primary structural source of Paper 15’s integers is $2T$ and its McKay / $\text{Cl}(0, 7)$ / $\text{Spin}(7)$ chain.

8.4 The Kerr connection

The electron’s Kerr–Newman ring radius is $a = \hbar/(2m_e c) = \xi_{\text{SM}}/2$ —exactly half the BPS healing length. Carter’s $g = 2$ [5] for the Kerr metric matches Dirac’s quantum result, and the charge radius $r_Q = \sqrt{\alpha} \ell_{\text{Pl}}$ is the geometric mean of topology and quantum gravity. The BPS vortex core regularises what would otherwise be a Kerr ring singularity.

8.5 The residual

The NLO prediction matches CODATA at 0.013%. The NNLO correction $(1 + \alpha/7 + \alpha^2/49 + \dots) = 1/(1 - \alpha/7)$ shifts G by only $\alpha^2/49 \approx 10^{-6}$, well below the current residual. The remaining 0.013% likely reflects the running of α between the Thomson limit and the gravitational-crossover scale, or experimental uncertainty in G itself (the CODATA central value has shifted by $\sim 0.04\%$ between adjustments).

9 Lagrangian context: the result as $\mathcal{L}_5$ of the minimal three-field NWT theory

The gravitational hierarchy examined in this paper fits as one slice of a larger structural picture developed in a companion manuscript. A proposed minimal relativistic Lagrangian whose torus-knot solitons are intended to reproduce NWT’s observational footprint is a three-field theory on $\mathbb{R}^{1,3}$ (or, in the GUT-extended UV completion, on a Kaluza–Klein space with internal $S^3/2I$):

$$\mathcal{L}_{\text{NWT}} = |D_\mu \psi|^2 - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{\lambda}{4} (|\psi|^2 - v^2)^2 + \frac{f_\pi^2}{2} (\partial_\mu n^a)^2 - \frac{1}{4e_{\text{Sk}}^2} (\partial_\mu n^a \partial_\nu n^b - \partial_\nu n^a \partial_\mu n^b)^2 + \theta Q_H[n]. \quad (15)$$

The fields are a complex scalar condensate ψ , an abelian gauge field A_μ , and an S^2 -valued Faddeev unit field n^a (which can be written as $n^a = \psi^\dagger \sigma^a \psi / (\psi^\dagger \psi)$ in the CP^1 extension). The couplings λ , f_π , e_{Sk} , θ are fixed at the BPS critical point $\lambda = e^2/2$, the Derrick equilibrium $\kappa = R/a_0 = \pi^2$, and the CPT-invariant $\theta = 0$ or π . No free fit parameters beyond the overall mass scale $v = m_e$.

The full structural programme decomposes into five layers, labelled $\mathcal{L}_1$ through $\mathcal{L}_5$, of which the present paper is the fifth:

- $\mathcal{L}_1$: minimal field content (ψ, A_μ, n^a) , with the $\text{Spin}(7)/\text{Cl}(0, 7)/2T$ structure of §8.2–3 emerging on the moduli space of torus-knot solitons, not as a fundamental gauge symmetry.
- $\mathcal{L}_2$: BPS abelian Higgs sector $(|D\psi|^2 + F^2 + V)$ at $\lambda = e^2/2$. Bogomolny bound saturated; line tension $\mu_{\text{BPS}} = 2\pi v^2$ verified to $\sim 10^{-3}\%$.
- $\mathcal{L}_3$: Skyrme–Faddeev quartic + Hopf θ -term, ensuring stable finite-size knotted solitons via Derrick scaling and locking the topological sector to $Q_H = p \cdot m$ for a $T(p, q)$ torus knot with phase winding m .

- $\mathcal{L}_4$: Paper 6 mass formula for 24 hadrons and leptons (1.06% median deviation, no fit parameters beyond the anchor m_e) is obtained from $\mathcal{L}_2$'s line tension evaluated on $\mathcal{L}_3$'s torus-knot solitons via the Kelvin–Saffman thin-ring approximation.
- $\mathcal{L}_5$: *this paper*. The gravitational hierarchy $m_e/m_{\text{Pl}} = (8/7)(1 + \alpha/7)\alpha^{21/2}$ is interpreted as arising from the one-loop Casimir of $\mathcal{L}_2$ on $S^3/2I$, with the $T(2,3)$ trefoil sector selected by $\mathcal{L}_3$'s Hopf θ -term.

Each factor in the gravitational formula can be associated with a proposed origin in Eq. 15. Table 6 summarises the mapping; the attributions are best read as a coherent map of the model's internal logic rather than as a completed microscopic derivation (for which see §8.2 on what is rigorous versus structurally grounded).

Factor	Lagrangian origin	Reference
8/7	$\mathcal{L}_1$ field content: dim(Spin(7) spinor)/ dim(Spin(7) vector) matter / gauge propagator ratio	§8.2, §3
$\alpha^{21/2}$	$\mathcal{L}_2$ gauge sector: product over 21 edges of Eulerian circuit on K_7 , each contributing $\sqrt{\alpha}$	§8.2
$(1 + \alpha/7)$	$\mathcal{L}_2$ one-loop NLO: seven K_7 vertex self-energies, each of order $\alpha/49$	§3.2
θQ_H lock	$\mathcal{L}_3$ Hopf θ -term: selects the $(2,3)$ trefoil sector via the $2I$ surgery equivalence	§8.2

Table 6: Attribution of each factor in $m_e/m_{\text{Pl}} = (8/7)(1 + \alpha/7)\alpha^{21/2}$ to a specific term in the minimal NWT Lagrangian (Eq. 15). Companion Paper 16 [3] develops the full $\mathcal{L}_1$ – $\mathcal{L}_5$ chain.

The forthcoming Paper 16 [3] assembles $\mathcal{L}_1$ – $\mathcal{L}_5$ into a single coherent treatment, including the $SO(10)$ UV completion at $E_{\text{GUT}} \approx 7.4 \times 10^{15}$ GeV (which adds 33 heavy gauge bosons orthogonal to the present paper's emergent Spin(7) structure), and three concrete falsifiers: Hyper-Kamiokande proton-decay reach $\tau(p \rightarrow e^+\pi^0) \sim 1.4 \times 10^{35}$ yr, non-MSSM coupling unification at $\alpha_{\text{GUT}} = 1/40$, and a GUT phase-transition gravitational-wave signature near 7.4 GHz. The present paper provides $\mathcal{L}_5$.

10 Conclusion

Within the accounting developed here, the trefoil knot organises a substantial portion of the framework's output:

1. Its **crossings** generate $\mathfrak{su}(3)$ and the fine structure constant α (Papers 8, 11, 13–14).

2. Its **Dehn surgery** produces the Poincaré sphere $S^3/2I$ and the icosahedral group A_5 , giving nuclear saturation $k_{\text{sat}} = 5$ (Paper 17) and the cinquefoil eigenvalue $q^2 = 25$ in the α formula (Paper 14).
3. Its **Poincaré spectrum** $\lambda_1 = 168 = |\text{Irr}(2T)| \times |2T| = 7 \times 24$ (McKay correspondence, extended E_6 Dynkin) supplies the integers entering the gravitational hierarchy $m_e/m_{\text{Pl}} = (8/7)(1+\alpha/7)\alpha^{21/2}$ (this paper).
4. Its Spin(7) **embedding** $2T \subset \text{Spin}(7)$ via the octonion Clifford algebra $\text{Cl}(0,7)$ provides an explicit matrix realisation; the three integers (7, 8, 21) in the gravitational hierarchy formula are the vector, spinor, and adjoint dimensions of $\text{Spin}(7) = B_3$ (this paper, Sec. 8.2).

The predicted gravitational constant agrees numerically with CODATA at the 0.013% level. Within the dimensionless content of the framework, no fit parameter remains (modulo the structural-vs-dynamical caveats of §8.2 and §7): the one-knot accounting reduces the Standard Model plus gravity from ~ 21 dimensionless parameters to zero.

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The spectral connection $\lambda_1 = 168 = 8 \times 21$ was discovered during an exploratory session on gravitational self-consistency and Kerr naked singularities, following the publication of Paper 14. The NLO pattern $(1 + \alpha/C_A)$ was identified by systematic stress-testing of the leading-order formula against alternative integer decompositions.

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All analysis code, simulation scripts (BPS profile solver, Poincaré eigenvalue computation, holonomy integrator, crossing algebra, knot selection scan), and the L^AT_EX source for this manuscript are available at <https://github.com/JimGalasyn/null-worldtube>.

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